1.R = R1 + R2 + R3

R= 20 + 30 + 50

R = 100

2.I=125\100

1.25A

3. I1 = I2 = I3 = 1.25A

4.Ohm's law (V = I *R) for each individual resistor.

- **20 Ohm Resistor:**

V1 = 1.25* 20 = 25V

- **30 Ohm Resistor:**

V2 = 1.25A * 30 = 37.5V

- **50 Ohm Resistor:**

V3 = 1.25 A * 50 = 62.5V

5. Use Watt's law P = V * I

- **20:**

P1 = 25 V*1.25A = 31.25W

- **$30:**

P2 = 37.5V* 1.25A} = 46.87 W

- **50:**

P3 = 62.5V *1.25 A} = 78.125W

PART B

1. 1/R = 1/R1 + 1/R2 + 1/R3

1/R = 1/20 + 1/100 + 1/50

1/R = 8/100

R = 100/8

R = 12.5

2. I=125/12.5

10

3. Ohm's law (I = V / R)

for each individual resistor.

20 Ohm Resistor:

I1 = 125 / 20 = 6.25A

100 Ohm Resistor:

I2 = 125 / 100 = 1.25A

50 Ohm Resistor:

I3 = 125 / 50 = 2.5A

4. V1 = V2 = V3 = 125V

5. Use Watt's law P = V * I

20 Ohm Resistor:

P1 = 125V * 6.25A =

781.25W

100 Ohm Resistor:

P2 = 125V * 1.25A =

156.25W

50 Ohm Resistor:

P3 = 125V * 2.5A =

312.5W